computing and optimisation solutions into a single cloud service. You can write code and run it on the quantum hardware of your choice. QCI, IonQ, and Quantinuum are some of those who have partnered with Azure. NASA's Jet Propulsion Lab used Azure Quantum to develop an optimisation solution that helped reduce scheduling times from hours to minutes. Ford and Microsoft teamed up to use quantum methods to figure out how to reduce traffic congestion in Seattle.

Google's Cirq is a Python software library for writing, manipulating, and optimising quantum circuits, and then running them on quantum computers and quantum simulators. Cirq provides useful abstractions for dealing with NISQ computers, where details of the hardware are vital to achieving useful results.

Amazon Braket provides customers access to quantum computing technologies from multiple quantum hardware providers, including D-Wave, IonQ, and Xanadu. It also offers a consistent set of development tools for you to build and test quantum computing projects.

Prof. Prabhakar adds that most of the software is built around an open source effort. "These are still early days, and a collaborative approach is likely to yield higher dividends. However, there is a danger of having too many alternatives for a beginner to comprehend. It helps that most platforms rely on Python libraries, making them amenable to a quick introduction," he says.

Applications galore

Although quantum computing is quite a few years away from becoming mainstream, it is time for companies to start exploring what it can do for them. Liscouski lists out a few key industrial applications, for us to understand the potential of quantum computing to transform businesses:

Supply chain and retail:

- Quantum computing enables users to improve multi-level replenishment throughout the entire supply chain from the manufacturer to the distribution centres, right down to local stores.
- Users can accelerate warehouse performance with highly optimised pick order and path routing, while accommodating the use of more autonomous robots (as well as the growing data complexity that comes with them).
- Quantum computing can perform inter-modal optimisations (for example, across long-haul and local surface freight) that would be very challenging for classical computers.

Cybersecurity:

- Cybersecurity is a perfect application for quantum computing because of the complexity of unstructured data that must be repetitively and efficiently analysed to prevent threats and respond quickly to intrusions.
- Quantum computing enables the real-time investigation of network intrusion to disrupt and prevent data exfiltration incidents.
- It also has the ability to detect patterns of divergent network traffic.

Quantum sensing:

- Quantum light detection and ranging (lidar) will extend the capabilities of existing lidar to provide enhanced data acquisition and analysis.
- Quantum sensing is also applicable in the medical imaging field; imagine being able to see below the skin without high power such as X-ray.

Life sciences and pharmaceutical:

- Quantum computers can assist in a wide range of application areas where understanding interrelatedness of common attributes is key to discovery. For example, identifying patient attributes and subpopulations in clinical trials.
- Quantum computing can help discover heterogeneous networks of proteins and genes implicated in human biological function and disease states or in viral or microbial life cycles.
- It can also analyse networks to compare molecular substructures and analyse the larger molecules necessary in pharmaceutical research.

"The early adopters are financial institutions seeking an advantage in wealth management, logistics companies attempting to reduce costs and improve efficiencies, and pharma companies that need to tackle the quantum chemistry of their new molecules. We also expect telecom and data security companies to be watchful of any breakthroughs in cryptography,

With simulators available, who needs quantum computers?

In 2019, Google researchers claimed they had achieved quantum supremacy when their quantum computer Sycamore performed an abstruse calculation, which would take a supercomputer 10,000 years to solve, in 200 seconds. Recently, a group of scientists in China created waves by completing the same computation in a few hours with ordinary processors. They recast the problem as a 3D mathematical array called a tensor network, and ran the simulation by multiplying all the tensors.

Quantum simulators are software programs that allow you to use a classical computer to run quantum circuits as if they were being run on a quantum computer. There are several simulators like Intel-QS and NVIDIA cuQuantum. And the question often arises as to why we need quantum computers when these simulators are there. While simulators can probably solve some problems, solving actual quantum problems using them would require an unmanageably high amount of computing resources. That said, simulators can be helpful in developing quantum computing applications and algorithms, as they can help debug code and test near-term systems.